

熱機械変動下における編隊飛行  
のための多センサ相対航法  
**Multi-Sensor Relative Navigation for  
Formation Flying Under Thermo-Mechanical  
Variations**

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# Background & Research Objectives

## Enhancement of Space Observation Capabilities

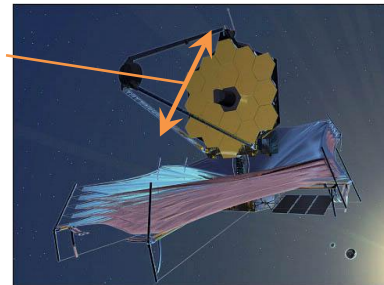
- To enhance the resolution of astronomical and Earth observations
  - Necessity to have telescopes with **longer focal lengths or larger apertures**.
  - Difficulty to enlarge them with a single spacecraft.



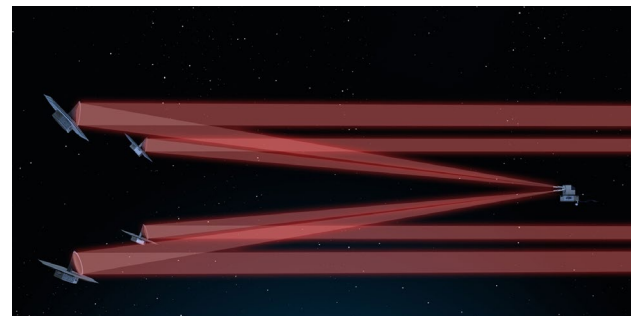
Rise of **Formation Flying (FF) missions**

- FF missions surpass the size limitations of the telescopes
  - Enable **a substantial improvement in observational resolution**.

Aperture:  
6.5m



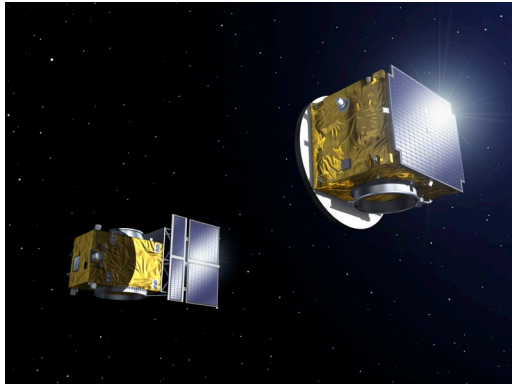
James Webb Space Telescope [1]



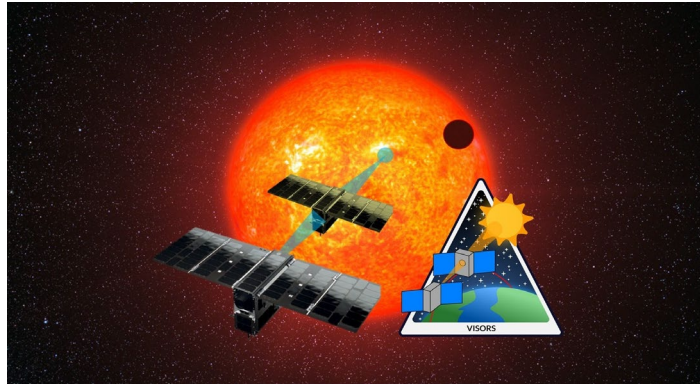
LIFE space mission [2]

## Formation Flying: New Way to Realize Large Effective Baselines

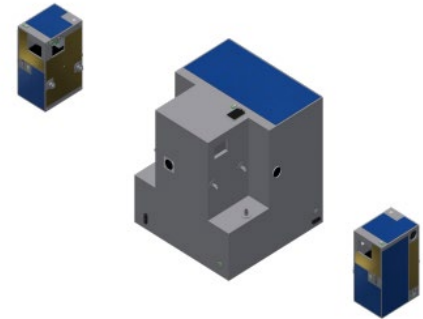
- Recent and planned missions already reflect this trend
  - PROBA-3 demonstrated millimeter-level formation control, while concepts such as VISORS and SEIRIOS require even tighter relative geometry.
- FF is becoming a core mission-enabling technology rather than a secondary supporting capability.



PROBA-3 (ESA) [3]



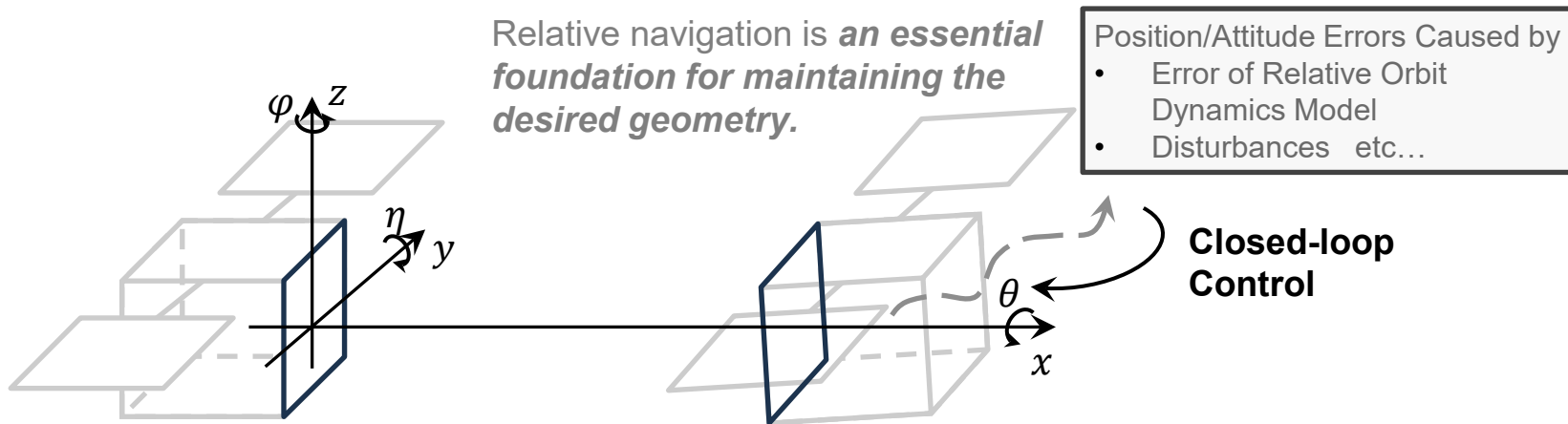
VISORS (USA) [4]



SEIRIOS

## Precise Relative Navigation Is Required to Maintain FF Geometry

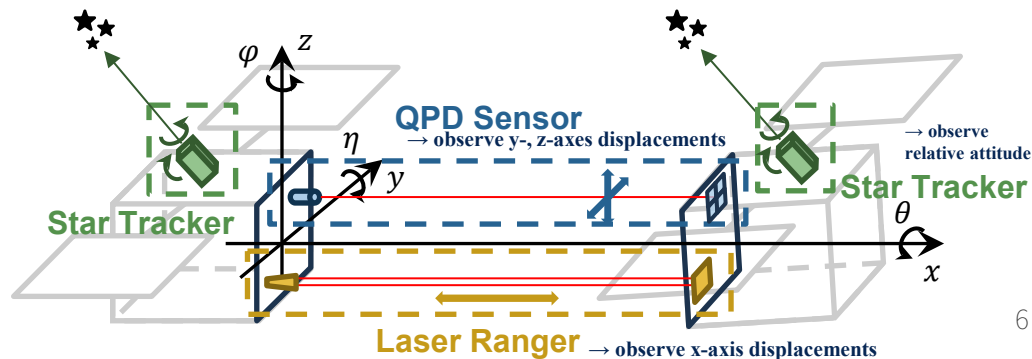
- FF must maintain a precisely-defined relative geometry between spacecraft — mission performance depends on it.
- **The relative position and attitude must therefore be estimated stably and precisely on orbit.**



## Robust 6-DoF Relative Navigation Requires Multi-Sensor Fusion

- High-precision 6-DoF relative navigation requires multiple sensing modalities.
  - Because each sensor is most sensitive to only a subset of the relative state, robust 6-DoF estimation must rely on a fusion architecture that leverages their complementary observability.

Although multi-sensor fusion can achieve high precision, its actual performance depends critically on **the fidelity of the measurement models**.

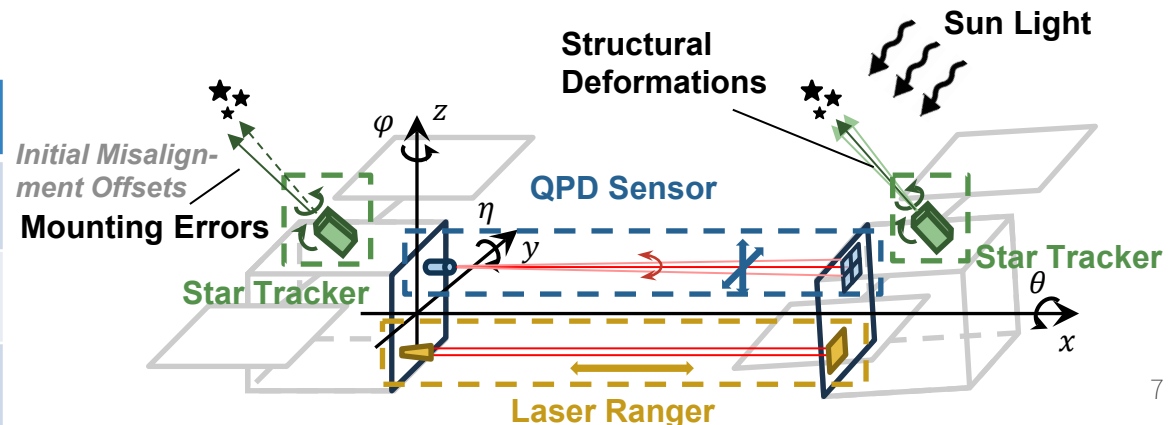


# Measurement-Model Errors as a Dominant Limitation for Precise FF

- Precise FF navigation performance depends critically on **the fidelity of the measurement models**.
  - Small mounting errors and structural deformations appear as changes in sensor extrinsics, and introduce persistent geometric biases that can dominate the estimation error budget.
- The key challenge is that **ground-calibrated measurement models cannot be assumed to remain valid on orbit**.

Thermal-distortion-driven misalignments in flight

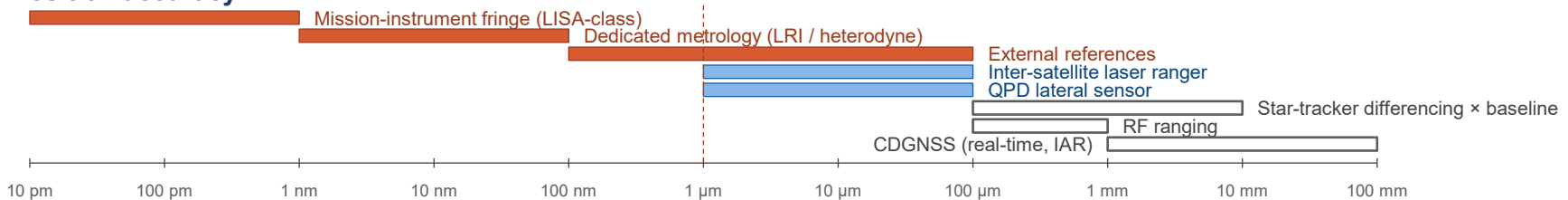
| Spacecraft     | $\Delta x$                            | $\Delta\theta$ (arcsec)           | Env-driven $\Delta T$                |
|----------------|---------------------------------------|-----------------------------------|--------------------------------------|
| Hubble [5]     | $\leq 10 \mu\text{m} / 4.9 \text{ m}$ | $\sim 1$                          | $\pm 20^\circ\text{C}$ day/night     |
| GOES [6]       | —                                     | $< 10$ (30 worst) $\mu\text{rad}$ | $\pm 75^\circ\text{C}$ seasonal      |
| Sentinel-2 [7] | —                                     | $3 \mu\text{rad}$                 | $\pm \text{several } ^\circ\text{C}$ |



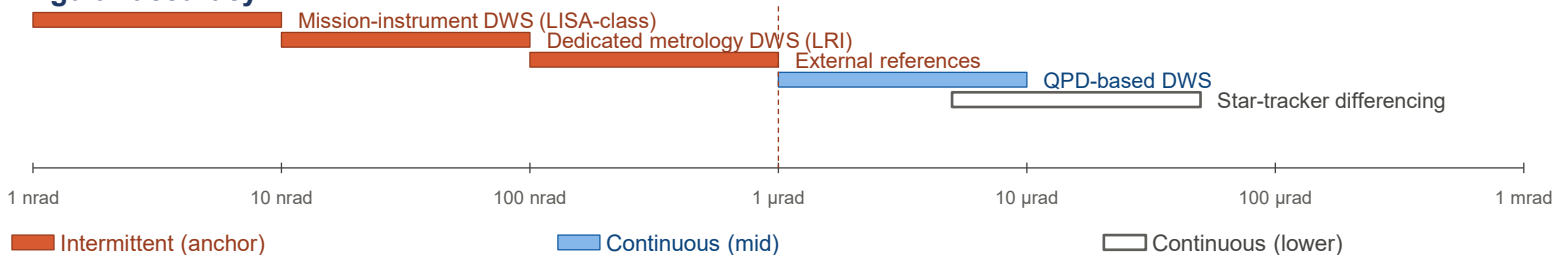
# Sensor Accuracy Hierarchy: Intermittent References as Calibration Anchors

Sensors differ in accuracy and availability — the most accurate are intermittent, but they can anchor lower-tier sensors.

## Position accuracy



## Angular accuracy

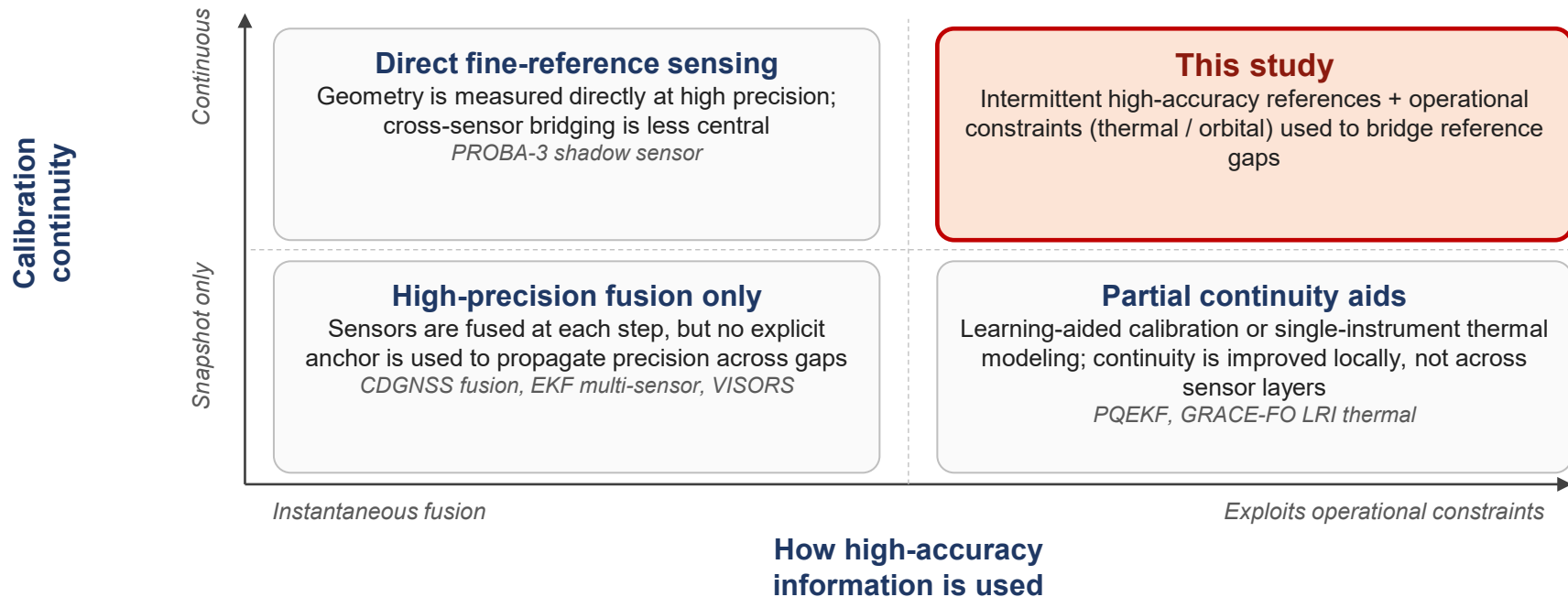


### KEY CHALLENGE

→ Anchor lower-tier sensor calibration to intermittent high-accuracy references.

# Research Map of Prior Studies

How are high-accuracy references used, and how is precision sustained between them?



- No prior work simultaneously uses intermittent high-accuracy references and operational constraints to continuously calibrate lower-layer sensors across reference gaps.

## Research Gap and Proposed Contribution

### THIS STUDY

#### **Reference-triggered onboard residual model learning**

Intermittent high-precision references update reusable residual models for onboard estimators. Batch regression, neural networks, and recursive updates are integrated to retain accuracy across reference gaps.

#### 1 OBJECTIVE

##### **On-orbit data adaptation**

Use intermittent high-precision references to calibrate on-board estimators and preserve accuracy when references are unavailable.

#### 2 FORMULATION

##### **Structured residual model**

Represent reference residuals as functions of physical, geometric, thermal, and orbital features using linear models or neural networks.

#### 3 METHODOLOGY

##### **Hybrid offline–online learning**

Build expressive residual priors with batch/NN models, and refine them on-board through reference-triggered recursive updates.

### → OUTCOME

Reusable onboard residual models that sustain high-accuracy navigation between intermittent references

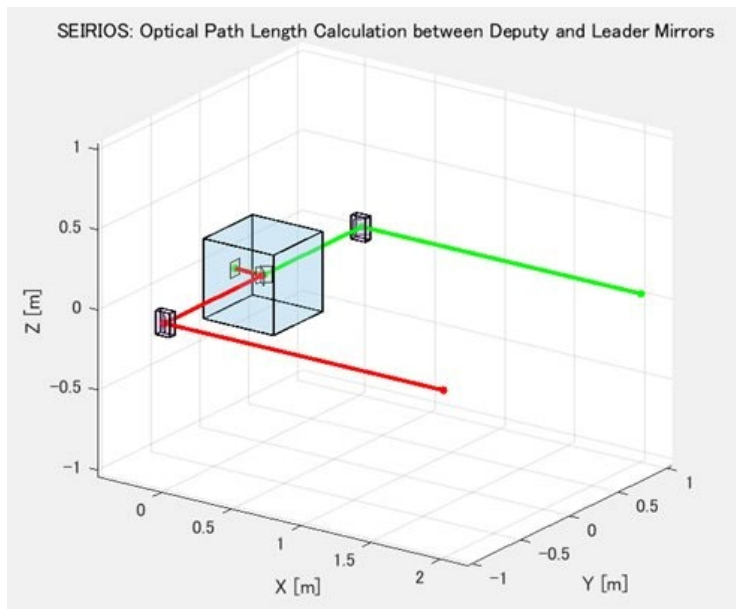
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# Methodology

## Case Study: SEIRIOS — A 3-Spacecraft IR Interferometer

Validation testbed: a 3-spacecraft IR interferometer where mission instruments provide intermittent high-precision references for relative navigation.



SEIRIOS: 1 chief satellite + 2 deputies; starlight relayed via deputy mirrors to the chief detector.

### MISSION OBSERVABLES (REFERENCE)

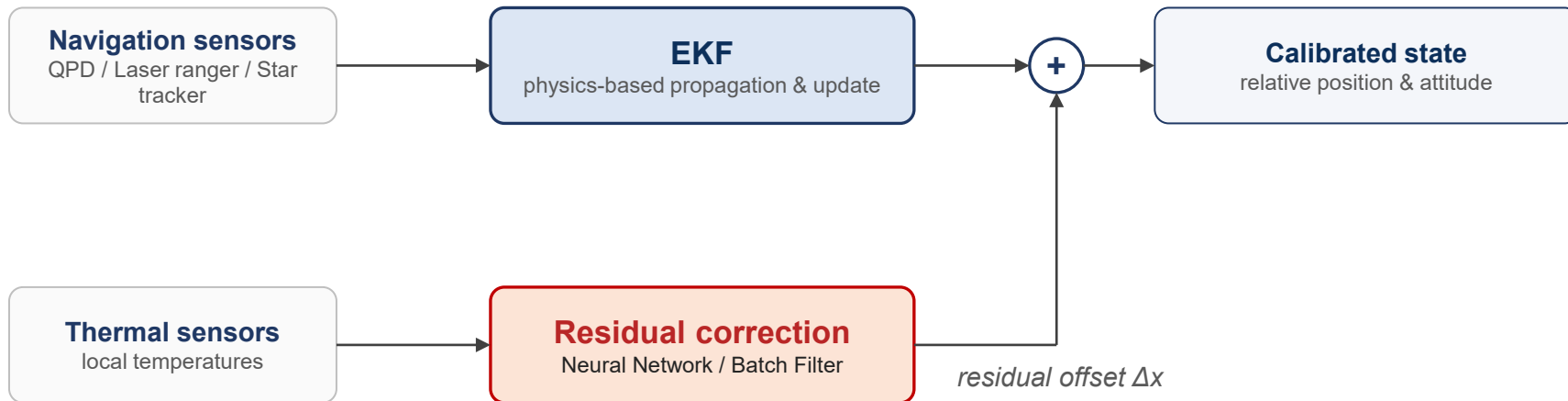
- $OPD = |B_2| - |B_1|$  : optical path difference between the two arms
- $\alpha, \beta$  : angular errors of the incoming beams
- $x_e, y_e$  : beam-spot positions on detector

### ESTIMATION TASK

Estimate  $OPD, \alpha, \beta, x_e, y_e$  from **continuous inter-satellite sensors** (QPD, laser ranger, star trackers) between fringe events.

# Learning-Augmented EKF: Architecture Overview

EKF provides the physics-consistent backbone; a residual model compensates what physics leaves behind.



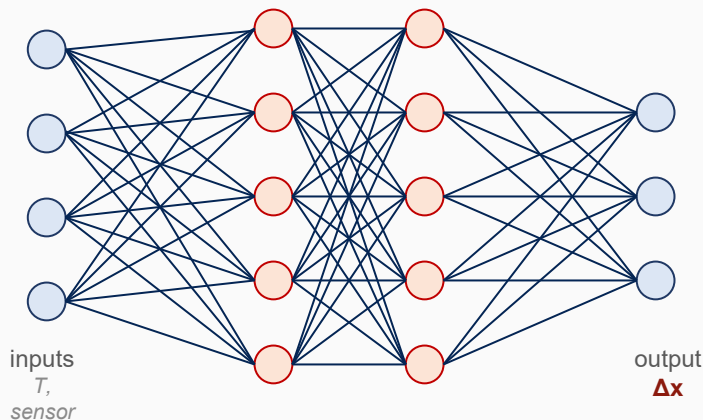
**State:  $x_{cal} = x_{EKF} + \Delta x$**

Residual correctors: batch/NN for offline learning, and reference-triggered RLS/VRF for onboard sequential adaptation.

## Variant 1: NN-Augmented LA-EKF

A neural network maps thermal and sensor context to a residual offset added to the EKF estimate.

### RESIDUAL MODEL = NEURAL NETWORK



### CHARACTERISTICS

- Flexible non-linear residual model
- Offline training on simulated thermal / sensor data
- Best when thermo-mechanical residuals are hard to model analytically
- Higher training cost; overfit control is required

$$\Delta x = g_{\theta}(s_k)$$

## Variant 2: Batch / Ridge Regression

A structured polynomial model maps thermal features to the residual offset and is identified by ridge least squares.

### RESIDUAL MODEL = POLYNOMIAL BASIS

Polynomial features of thermal channel:

$$\boldsymbol{\varphi}(T) = [1, (T-c), (T-c)^2]$$

Residual offset:

$$\Delta x = \boldsymbol{\Phi}(T) \mathbf{B}$$

Identified by ridge regression:

$$\hat{\mathbf{B}} = \operatorname{argmin} \|\boldsymbol{\Phi} \mathbf{B} - \Delta \mathbf{Y}\|^2 + \lambda \|\mathbf{B}\|^2$$

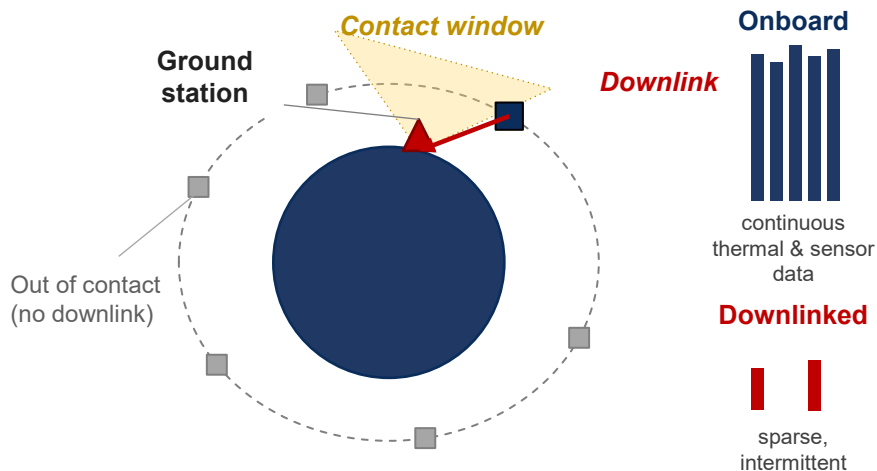
### CHARACTERISTICS

- Low-capacity structured model
- Closed-form ridge identification
- Fast to train and easy to interpret
- Limited extrapolation beyond the basis / training span

Trade-off: Batch is fast and lightweight, whereas NN is heavier but usually extrapolates better.

# Toward Onboard Adaptation: Reference-Triggered RLS / VRF

Ground contact is short and infrequent: only fragments of the onboard data ever reach the ground, while the satellite itself records continuously. This asymmetry motivates dynamic on-orbit calibration.



## WHY ONBOARD ADAPTATION?

- **Limited downlink**  
Ground retraining sees only a fraction of the on-orbit data — sparse and intermittent.
- **Rich onboard data**  
Thermal and sensor histories are recorded continuously across every orbit.
- **Idea: close the loop on orbit**  
Use rare reference signals to update the residual model in flight, instead of waiting for ground contact.

**Bandwidth asymmetry: dynamic onboard calibration uses what is abundant on orbit, instead of waiting for what is scarce on the ground.**

## Recursive Update: Fixed-Forgetting RLS and VRF-RLS

Both modes share the same recursive update; VRF replaces the constant  $\lambda$  with an innovation-driven  $\lambda_k$ , following Bruce, Goel & Bernstein (2020).

### FIXED-FORGETTING RLS

At a reference event ( $y_k = 1$ ):

$$\mathbf{e}_k = \mathbf{r}_k - \boldsymbol{\varphi}_k^T \boldsymbol{\theta}_k$$

$$\mathbf{K}_k = \mathbf{P}_k \boldsymbol{\varphi}_k / (\lambda + \boldsymbol{\varphi}_k^T \mathbf{P}_k \boldsymbol{\varphi}_k)$$

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k + \mathbf{K}_k \mathbf{e}_k$$

$$\mathbf{P}_{k+1} = \lambda^{-1} (\mathbf{I} - \mathbf{K}_k \boldsymbol{\varphi}_k^T) \mathbf{P}_k$$

Constant  $\lambda \approx 0.999$ . Suited to slow drift.

### VARIABLE-RATE FORGETTING (VRF-RLS)

EWMA innovation variance:

$$\sigma_{\mathbf{e},k}^2 = (1 - \mu) \sigma_{\mathbf{e},k-1}^2 + \mu \mathbf{e}_k^2$$

Predicted innovation variance:

$$\alpha_k = \boldsymbol{\varphi}_k^T \mathbf{P}_k \boldsymbol{\varphi}_k$$

Normalized squared innovation:

$$\zeta_k = \min (\mathbf{e}_k^2 / (\sigma_{\mathbf{e},k}^2 + \alpha_k + \varepsilon), 1)$$

Time-varying forgetting:

$$\lambda_k = 1 - (1 - \lambda_{\min}) \zeta_k$$

Large  $\mathbf{e}_k \rightarrow$  small  $\lambda_k$  (fast adaptation); steady state  $\rightarrow \lambda_k \rightarrow 1$  (long memory). [9]

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# Results

# Simulation Setup: Thermal Histories from Multi-Node Analysis

On-orbit thermal histories of the three SEIRIOS spacecraft are generated using an open-source multi-node thermal simulator (training-data pipeline detailed on the next slide).

## THERMAL TOOL

### spacecraft-thermal-multi-node-analysis

Open-source multi-node thermal analysis [10]

*Outputs node-level on-orbit temperature histories*

## CONFIGURATION

### 3-spacecraft formation

- Sun-to-face angle swept from  $0^\circ$  to  $90^\circ$
- Per-node temperature histories fed to the residual model

## TRAIN / TEST SPLIT

### Sun-angle split

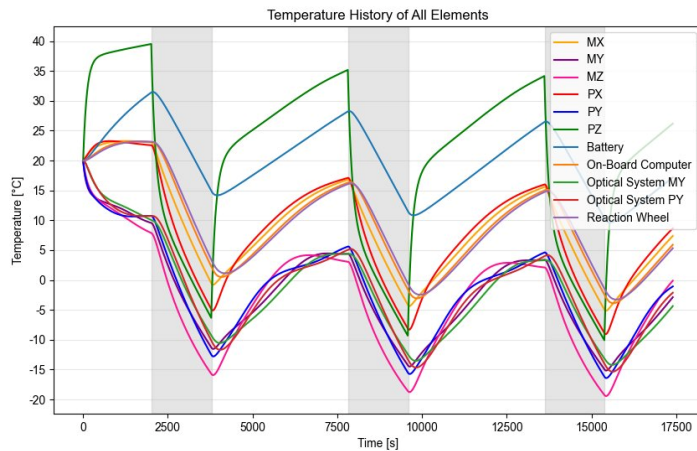
- Train: 5 angles
  - Validation / test: unseen angles
- Evaluates generalization across thermal histories*

## EVALUATION

- Same relative-motion profile and sensor-noise assumptions for all methods
- Compared methods: EKF baseline / LA-EKF (NN) / LA-EKF (Batch) / Proposed (distance-normalized NN)

# Training Data Pipeline: Residual $\leftrightarrow$ Temperature

Supervised target = (mission-truth) - (EKF estimate). Learn its dependence on



*Multi-orbit temperature histories per node (varies with sun angle & eclipse)*

## STEP 1 — GENERATE SCENARIOS

Multi-node thermal simulator generates **multi-orbit temperature histories** for **several sun-to-face angles**.

## STEP 2 — COLLECT RESIDUALS

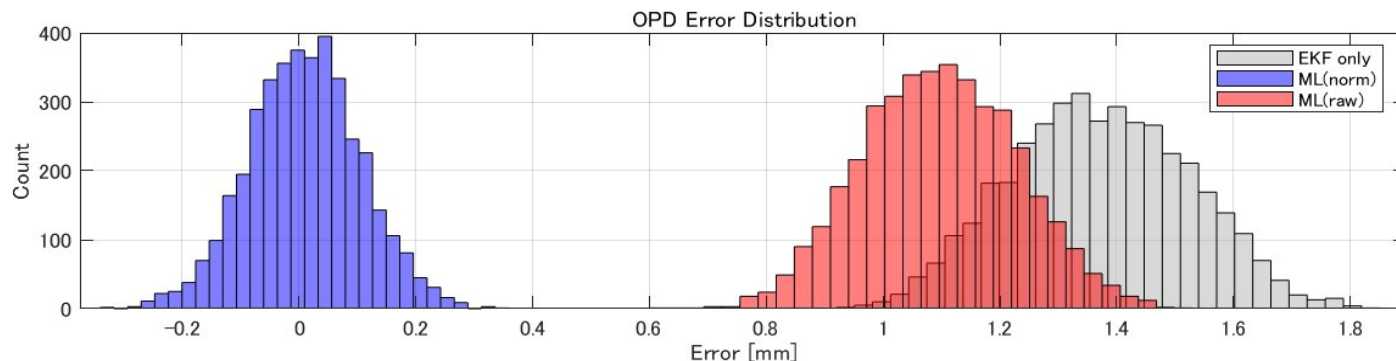
For each scenario, take residual = (mission instrument: OPD,  $\alpha$ ,  $\beta$ ) - (physics/geometry model + EKF).

## STEP 3 — LEARN ( $\Delta T \rightarrow$ RESIDUAL)

Pair  $\Delta T$  (from temperature sensors)  $\leftrightarrow$  residual. Train NN / Batch / RLS to map  $\Delta T \rightarrow$  residual.

## EKF Alone Cannot Compensate Thermo-Mechanical Bias

EKF baseline leaves a ~1.4 mm OPD bias; distance-normalizing the OPD target — not just adding an NN — removes it.



*OPD error distribution at sun-angle = 0° and baseline = 50 m. Narrower and more centered is better.*

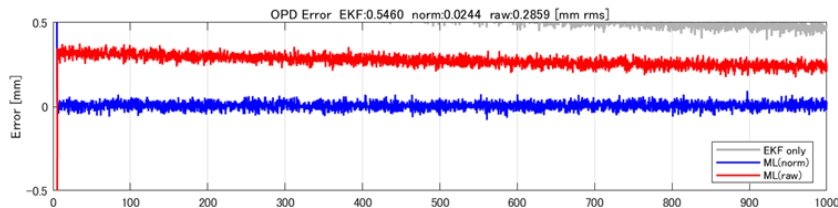
### FINDINGS

- EKF baseline: 1.38 mm RMS — thermo-mechanical bias is not modeled
- **LA-EKF (NN, raw OPD target): 1.10 mm RMS — variance is reduced, but a large bias remains**
- **Proposed (NN, distance-normalized OPD target): 0.10 mm RMS — bias is largely removed (~14× improvement)**

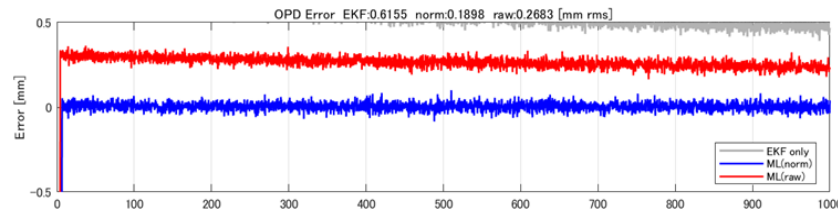
## Distance-Normalized NN Generalizes Across Thermal Histories

OPD error vs. time at three sun angles ( $d = 50$  m). Distance-normalization ( $\Phi_{\text{norm}}$ ) is what enables generalization.

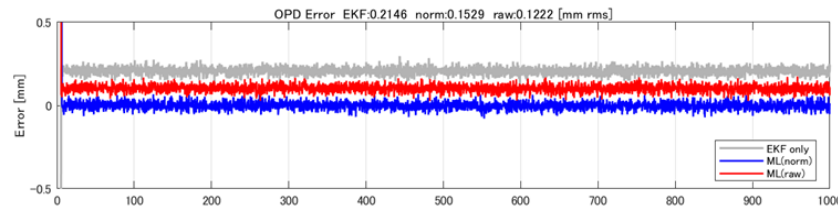
$\beta = 0^\circ$



$\beta = 45^\circ$



$\beta = 82^\circ$

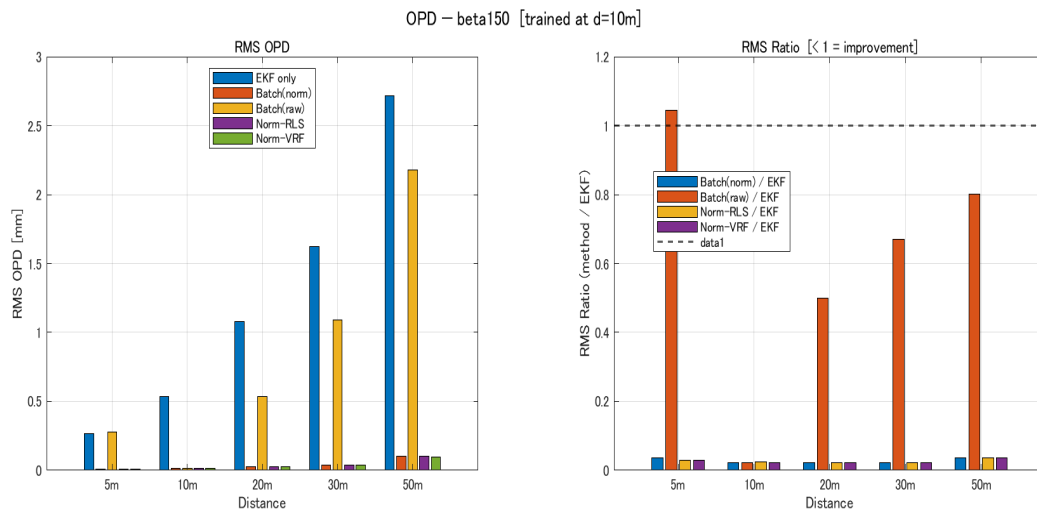


| RMS [mm]            | $\beta=0^\circ$ | $\beta=45^\circ$ | $\beta=82^\circ$ |
|---------------------|-----------------|------------------|------------------|
| EKF baseline        | 1.38            | 1.40             | 0.55             |
| NN raw target       | 1.10            | 1.08             | 0.44             |
| <b>Dist-norm NN</b> | <b>0.10</b>     | <b>0.24</b>      | <b>0.20</b>      |

## Onboard Story: NN Aside, Compare Batch vs RLS / VRF — No-Misalignment Baseline

Onboard pivot: NN aside, comparing offline Batch vs online RLS / VRF. With no sensor-mounting shift, all corrected methods agree (train d = 10 m, eval 5–50 m, 8 patterns).

OPD RMS [mm] (8-pattern avg.)



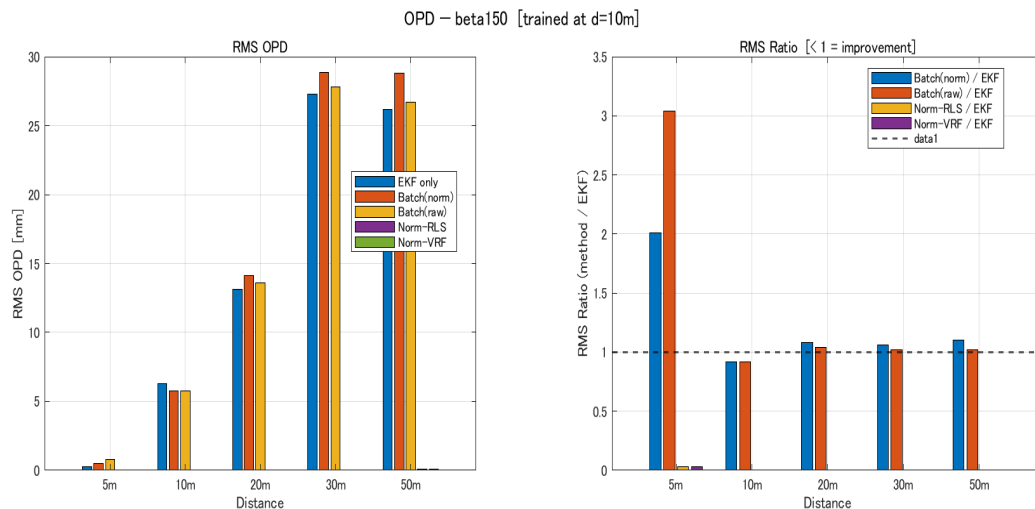
| d    | EKF   | Batch | RLS          | VRF          |
|------|-------|-------|--------------|--------------|
| 5 m  | 0.263 | 0.010 | <b>0.008</b> | <b>0.008</b> |
| 10 m | 0.537 | 0.012 | <b>0.013</b> | <b>0.012</b> |
| 20 m | 1.078 | 0.023 | <b>0.023</b> | <b>0.022</b> |
| 30 m | 1.624 | 0.037 | <b>0.036</b> | <b>0.036</b> |
| 50 m | 2.720 | 0.098 | <b>0.098</b> | <b>0.097</b> |

**Validation: Norm-RLS / VRF reach offline-batch accuracy across 5–50 m, despite training only at 10 m.**

## Sensor Pose Shift After Training: Batch Holds a Bias, RLS / VRF Recover It

Sensor mounting positions / attitudes at training time differ from those at inference time ( $\sigma_{\text{pos}} = 0.05$  mm,  $\sigma_{\text{att}} = 0.1$  mrad). Offline batch leaves a steady-state bias; only online RLS / VRF re-identify it on orbit.

OPD RMS [mm] (8-pattern avg.)



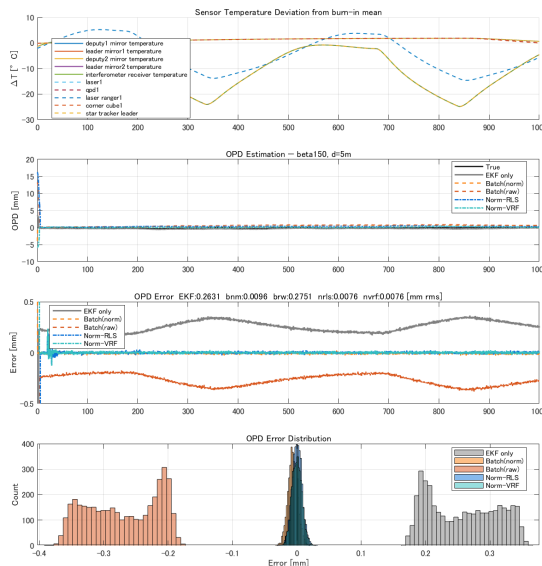
| d    | EKF   | Batch | RLS / VRF | Improvement |
|------|-------|-------|-----------|-------------|
| 5 m  | 0.255 | 0.512 | 0.008     | ~640×       |
| 10 m | 6.283 | 5.757 | 0.012     | ~480×       |
| 20 m | 13.10 | 14.15 | 0.023     | ~615×       |
| 30 m | 27.28 | 28.86 | 0.036     | ~802×       |
| 50 m | 26.20 | 28.81 | 0.099     | ~291×       |

Online adaptation is the unique enabler: 200–560× OPD-RMS reduction over offline batch when alignment shifts.

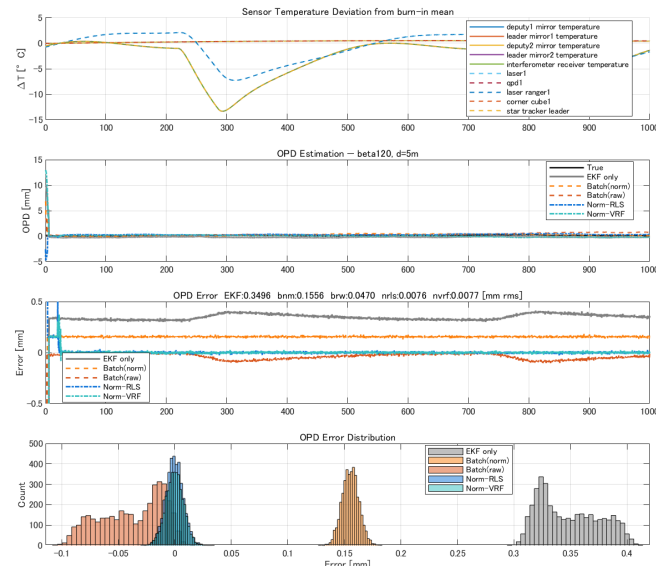
# Time-Series View: Online Convergence vs Offline Stagnation

Representative case at  $d = 50$  m. Without misalignment (left), all corrected methods stay tight on the truth. With misalignment (right), only Norm-RLS / VRF re-converge on orbit; offline batch tracks the wrong mapping.

## NO MISALIGNMENT ( $d = 50$ m)



## WITH MISALIGNMENT ( $d = 50$ m)

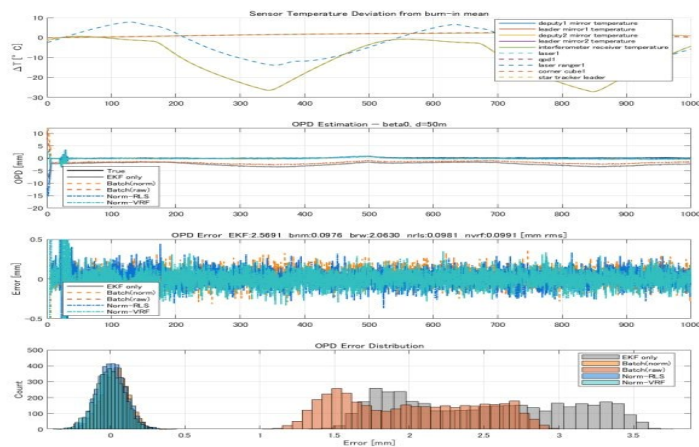


Norm-RLS / VRF re-identify the residual within tens of seconds; the gap to offline-batch is bounded by reference availability.

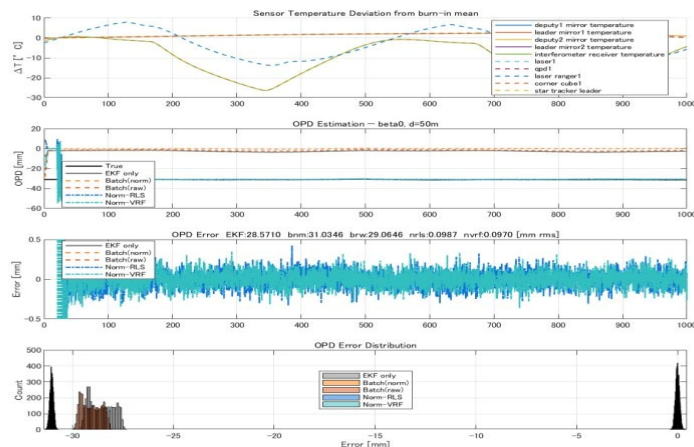
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## Future Work

# Toward Low-Cost Onboard Adaptation of Neural Residual Models

## NEXT STEP

### Low-cost recursive adaptation of neural residual models

Instead of retraining the full NN onboard, use the NN as a nonlinear feature extractor and update only a low-dimensional output layer or adapter using intermittent reference measurements.

#### 1 TECHNICAL APPROACH

##### **Recursive updates on low-dimensional NN layers**

Freeze the NN backbone and adapt only the last layer or adapter from intermittent reference residuals.

#### 2 RESEARCH QUESTIONS

##### **Adaptivity vs. stability**

Can NN residual models adapt to new on-orbit conditions while preserving EKF innovation consistency and memory across thermal cycles?

#### → TARGET

A reference-triggered LA-EKF in which NN-based residual priors are adapted through low-cost recursive updates while preserving EKF consistency across reference gaps.

# References

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8. K. Nakamura et al., spacecraft-thermal-multi-node-analysis, open-source software, 2025.
9. A. L. Bruce, A. Goel, and D. S. Bernstein, “Convergence and consistency of recursive least squares with variable-rate forgetting,” Automatica, vol. 119, p. 109052, 2020.
10. K. Nakamura, et al. spacecraft-thermal-multi-node-analysis from <https://pypi.org/project/spacecraft-thermal-multi-node-analysis/>

# Appendix

## Assumed Measurement Accuracy of Each Sensor

- The measurement accuracy of each sensor was defined as follows when simulating the relative position and orientation estimation algorithm:

**Table 1** Assumed measurement accuracy of each sensor in simulation.

| Sensor       | Output                 | Accuracy ( $1\sigma$ )                  | Notes                                                    |
|--------------|------------------------|-----------------------------------------|----------------------------------------------------------|
| QPD sensor   | beam-spot displacement | $\sigma_{\text{QPD}} = 20 \mu\text{m}$  | In-plane displacement orthogonal to nominal LoS.         |
| Laser ranger | Inter-satellite range  | $\sigma_{\text{LR}} = 10 \mu\text{m}$   | Two-way range with the coupling/misalignment correction. |
| Star tracker | attitude (Euler error) | $\sigma_{\text{ST}} = 2 \text{ arcsec}$ | Relative attitude from differenced absolute attitudes.   |